

**THE UNIVERSITY OF HONG KONG
FACULTY OF BUSINESS AND ECONOMICS**

MSBA7004 Operations Analytics

GENERAL INFORMATION

Instructor: Dr. Huiyin Ouyang
Email: oyhy@hku.hk
Office: KK 1321
Consultation times: TBD

Dr. Lin Chen
Email: linch@hku.hk
Office: KK 518
Consultation times: TBD

Tutors: Yifan Zhao, Yiran Zhang
Pre-requisites: MSBA7003 Quantitative Analysis Methods
Co-requisites: MSBA7002 Business Statistics
Mutually exclusive: N/A

Course Website: moodle.hku.hk
Other important details: TBD

COURSE DESCRIPTION

This course introduces the concepts of operations management and the application of analytic techniques in solving problems an organization faces in its operations.

An organization's ultimate success depends on how efficiently and effectively it executes its strategic goals. This requires its operational goals to be aligned with its strategic goals. A detailed understanding of the processes used to produce and deliver goods and/or provide services to customers is essential. Operations Management is a field which focuses on these aspects of an organization.

Operations of an organization concerns the management of the processes which convert inputs into outputs. Effective operations management involves managing people, equipment, and other resources. In the typical business organization the majority of the costs and controllable assets are managed by the operations function. Thus, this discipline—with the techniques, procedures and knowledge it encompasses—is a vital segment of business activity.

This course will provide students with the analytical tools needed to understand and articulate the impact of an organization's processes and the ability to continuously improve these processes.

COURSE OBJECTIVES

By introducing rigorous analytic methods and theories, this course demonstrates ways to apply structured thinking on loosely defined real-life operations processes. Upon successfully completing this course, students should be able to

1. Employ the fundamental concepts in process analysis,
2. Understand how to apply analytical models and theories in business processes,
3. Understand the concepts of process improvement, and
4. Apply an analytical framework in improving business processes.

Programme Learning Outcomes

PLO1: Acquisition and internalization of knowledge of the programme discipline			
PLO2: Application and integration of knowledge			
PLO3: Inculcating professionalism and leadership			
PLO4: Developing global outlook			
PLO5: Mastering communication skills			
COURSE LEARNING OUTCOMES			
Course Learning Outcomes			Aligned Programme Learning Outcomes
CLO1: Clearly identify and define a loosely structured operations process.			PLO1, 2, 3, 4
CLO2: Use analytical approaches to assess operations of a process.			PLO1, 2
CLO3: Apply process improvement tools to redesign an operations process.			PLO1, 2, 5
CLO4: Develop skills in designing and implementing an effective operations system.			PLO1, 2, 3, 5
COURSE TEACHING AND LEARNING ACTIVITIES			
Course Teaching and Learning Activities		Expected contact hour	Study Load (% of study)
T&L1. Lecture with interactive presentation		30	25%
T&L2. Assignments		36	30%
T&L3. Tutorial and e-forum discussions		24	20%
T&L4. Self-study		30	25%
Total		120	100%
Assessment Methods			
Brief Description (Optional)		Weight	Aligned Course Learning Outcomes
A1. Individual assignments	In-class participation + Attendance	30%	CLO1, 2, 3, 4
A2. Group assignment		10%	CLO1, 2, 3, 4
A3. Participation		10%	CLO1, 2, 3, 4
A4. Final exam		50%	CLO1, 2, 3, 4
Total		100%	
STANDARDS FOR ASSESSMENT			
Course Grade Descriptors			
A+, A, A-	• Demonstrate a strong understanding of all relevant knowledge • Handling questions professionally • High participation in discussions and volunteering answering/asking questions • Present arguments that have an element of originality • Achieve a standard of excellent performance in the exams with very accurate computation and very good analytical and problem solving skills • Excellent writing report and presentation		
B+, B, B-	• Demonstrate a good understanding of all relevant knowledge • Handling questions in a logical way		

	• Good participation in discussions • Present arguments that go beyond the lecture and textbook • Achieve a standard of good performance in the exams with accurate computation and good analytical and problem solving skills • Good writing report and presentation
C+, C, C-	• Demonstrate a basic understanding of the concepts involved • Fairly address questions as set • Some participation in discussions • Present arguments in a well-structure manner • Meet a standard of acceptable performance in the exams with reasonably accurate computation and acceptable analytical and problem solving skills • Acceptable writing report and presentation
D+, D	• Demonstrate a minimum understanding of the concepts involved • Barely address questions as set • Minimal or no participation in discussions • Present arguments in a marginally acceptable manner • Meet a standard of marginally acceptable performance in the exams with some errors in computation and barely adequate analytical and problem solving skills • Marginally acceptable writing report and presentation
F	• Demonstrate a poor understanding of the concepts involved • Unable or unwilling to handle questions • Minimal or no participation in discussions • Present arguments poorly • Fail to meet a standard of passing the exams with major errors in computation and inadequate analytical and problem solving skills • Poorly writing report and presentation
Assessment Rubrics for Written Assignments and Exams:	
A+, A, A-	• Demonstrate a strong understanding of all relevant knowledge • Present arguments that have an element of originality • Achieve a standard of excellent performance in the assessments with very accurate computation and very good analytical and problem solving skills • Excellent writing report and presentation
B+, B, B-	• Demonstrate a good understanding of all relevant knowledge • Present arguments that go beyond the lecture and textbook • Achieve a standard of good performance in the assessments with accurate computation and good analytical and problem solving skills • Good writing report and presentation

C+, C, C-	• Demonstrate a basic understanding of the concepts involved • Present arguments in a well-structure manner • Meet a standard of acceptable performance in the assessments with reasonably accurate computation and acceptable analytical and problem solving skills • Acceptable writing report and presentation
D+, D	• Demonstrate a minimum understanding of the concepts involved • Present arguments in a marginally acceptable manner • Meet a standard of marginally acceptable performance in the assessments with some errors in computation and barely adequate analytical and problem solving skills • Marginally acceptable writing report and presentation
F	• Demonstrate a poor understanding of the concepts involved • Present arguments poorly • Fail to meet a standard of passing the assessments with major errors in computation and inadequate analytical and problem solving skills • Poorly writing report and presentation
Assessment Rubrics for In-class Participation:	
A+, A, A-	• High participation in discussions • Always attend in-class discussions • Demonstrate a strong understanding of all relevant knowledge • Handling questions professionally • Present arguments that have an element of originality • Respect others and follow the class rules (no chatting and do not use cell phone)
B+, B, B-	• Good participation in discussions • Often attend the in-class discussions • Demonstrate a good understanding of all relevant knowledge • Handling questions in a logical way • Present arguments that go beyond the lecture and textbook • Respect others and follow the class rules (no chatting and do not use cell phone)
C+, C, C-	• Some participation in discussions • Sometimes attend the in-class discussions • Demonstrate a basic understanding of the concepts involved • Fairly address questions as set • Present arguments in a well-structure manner • Respect others and follow the class rules (no chatting and do not use cell phone)
D+, D	• Minimal or no participation in discussions • Rarely attend the in-class discussions • Demonstrate a minimum understanding of the concepts involved

	• Barely address questions as set • Present arguments in a marginally acceptable manner • Respect others and follow the class rules (no chatting and do not use cell phone)
F	• Minimal or no participation in discussions • Almost never attend the tutorials and in-class discussions • Demonstrate a poor understanding of the concepts involved • Unable or unwilling to handle questions • Present arguments poorly • Behave poorly in class (often chatting with others, using cell phones, or being late)

COURSE CONTENT AND TENTATIVE TEACHING SCHEDULE

Class	Topic
1	Introduction to Operations Analytics: What are Business Operations? Process analysis: Capacity analysis
2	Process analysis: Capacity analysis, continued
3	Process analysis: Case study Process analysis: Inventory and Little's Law <u>Due Individual Assignment 1 by start of Class 3: Case Analysis</u>
4	Variability analysis: Uncertainty in operations and queueing models
5	Process analysis: <i>Shouldice hospital case</i> Process analysis: Multi units, Project management and critical path method <u>Due Individual Assignment 2 by start of Class 5: Shouldice hospital case</u>
6	Inventory analysis: Economic Order Quantity, demand uncertainty, lead time, and safety stock Inventory analysis: Continuous vs. Periodic review
7	Inventory analysis: Demand uncertainty and <i>Newsvendor</i> Case: Dual sourcing management
8	Inventory management case: Dual sourcing game <u>Due Group Assignment by start of Class 8: Dual-sourcing strategy</u>
9	Supply chain analysis: Bullwhip effect and incentive conflict <u>Due Individual Assignment 3 after Class 9: Inventory</u>
10	Revenue analysis: Selling to a <i>Newsvendor</i> Revenue management <u>Due Individual Assignment 4 after Class 10</u>

REQUIRED/RECOMMENDED READINGS & ONLINE MATERIALS (e.g. journals, textbooks, website addresses etc.)

Required Reading Materials:

1. Class notes and other materials posted on Moodle or handed out in class.
2. "Kristen's Cookie Co. (A)," Harvard Business Review Case 608037-PDF-ENG.
3. Egg waffle shop. Asian Case Center (Pending)
4. "Shouldice Hospital" (From textbook *Operations and Supply Chain Management*, Jacobs, F. Robert and Chase, Richard B., McGraw-Hill/Irwin) (15th edition: page 127 - 129)
5. Dual Sourcing Planning (Kellogg Business Case)

Recommended Reading Materials:

6. Anupindi, R., S. Chopra, S. D. Deshmukh, J. A. Van Mieghem, and E. Zemel, *Managing Business Process Flow*, (3rd Edition, Pearson New International Edition) Previous editions are also acceptable.
7. Jacobs, F. Robert and Chase, Richard B, *Operations and Supply Chain Management*, (14th Edition, McGraw-Hill/Irwin) Previous editions are also acceptable.
8. Cachon, G. and Terwiesch, C., 2023. *Matching supply with demand*. (5th Edition, McGraw-Hill).

MEANS/PROCESSES FOR STUDENT FEEDBACK ON COURSE

- ☒ Conducting mid-module survey in addition to SETL at the end of the module
- ☐ Online response via Moodle site
- ☐ Others: _____ (please specify)

COURSE POLICY (e.g. plagiarism, academic honesty, attendance, etc.)

An orderly learning environment is extremely important for this course. Disruptive behaviors are inconsiderate to other students as well as to the instructor, and are absolutely unacceptable. Talking during lectures, arriving to class late, and any other disruptions of mobile devices are not allowed; students who are responsible for any of these actions will be subject to academic penalty and will be asked to leave the classroom.

Any dishonesty—such as cheating, false representation, plagiarism, etc.—that comes to my attention will result in an F in the course.

Academic dishonesty includes cheating, plagiarism, unauthorized collaboration, falsifying academic records, and any act designed to avoid participating honestly in the learning process. Scholastic dishonesty also includes, but is not limited to, providing false or misleading information to receive a postponement or an extension on an exam or other assignment. The responsibilities of both students and faculty with regard to scholastic dishonesty are described in detail in the Disciplinary Committee Regulations (<http://www4.hku.hk/pubunit/calendar/2016-2017/a/c/32-2016-2017/400-disciplinary-committee-regulations>). By teaching this course, I have agreed to observe all of the faculty responsibilities described in that document. By enrolling in this class, you have agreed to observe all of the student responsibilities described in that document. If the application of that policy statement to this class and its assignments is unclear in any way, it is your responsibility to ask me for clarification.

ADDITIONAL COURSE INFORMATION (e.g. e-learning platforms & materials, penalty for late assignments, etc.)

Moodle system will provide a platform for discussion and knowledge exchange. Late assignments will not be accepted and will not receive any points.